Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 1 — solutions

September 11, 2026

Exercise 1. *Determine the truth values (i.e., T or F) of the following propositions:*

F $19 - 4 = 12$ *if and only if* 3 *is a prime number.*

T *If* $1 + 1 = 5$*, then* $1 + 1 = 3$.

T *If the moon is a star, then so is the sun.*

F *If* 5 *is a prime number, then the earth is flat.*

F $0 > 1$ *if and only if* $2 > 1$.

F *Either Toronto is the capital of Canada or Hamburg is the capital of Germany.*

Exercise 2. *Construct a truth table for each of these compound propositions:*

1. $p \oplus (p \vee q)$

p	q	$p \vee q$	$p \oplus (p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	F

2. $p \wedge (q \oplus u)$

p	q	u	$q \oplus u$	$p \wedge (q \oplus u)$
T	T	T	F	F
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	F	F
F	T	F	T	F
F	F	T	T	F
F	F	F	F	F

3. $(p \wedge q) \oplus (p \wedge u)$

p	q	u	$p \wedge q$	$p \wedge u$	$(p \wedge q) \oplus (p \wedge u)$
T	T	T	T	T	F
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

4. $(p \leftrightarrow q) \oplus (p \rightarrow q)$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$(p \leftrightarrow q) \oplus (p \rightarrow q)$
T	T	T	T	F
T	F	F	F	F
F	T	F	T	T
F	F	T	T	F

Exercise 3. *Without using truth tables, show the following logical equivalences:*

1. $\neg p \leftrightarrow q \equiv p \leftrightarrow \neg q$

$$\begin{array}{rcll} \neg p \leftrightarrow q & \equiv & (\neg p \rightarrow q) \wedge (q \rightarrow \neg p) & \text{Definition} \\ & \equiv & (\neg q \rightarrow p) \wedge (p \rightarrow \neg q) & \text{Contrapositive} \\ & \equiv & p \leftrightarrow \neg q & \text{Definition} \end{array}$$

2. $p \oplus (q \wedge u) \not\equiv (p \oplus q) \wedge (p \oplus u)$.

Set $p = \mathrm{T}$, $q = \mathrm{F}$, and $u = \mathrm{T}$. Then $p \oplus (q \wedge u) = \mathrm{T} \oplus \mathrm{F} = \mathrm{T}$. On the other hand, $(p \oplus q) \wedge (p \oplus u) = (\mathrm{T} \oplus \mathrm{F}) \wedge (\mathrm{T} \oplus \mathrm{T}) = \mathrm{T} \wedge \mathrm{F} = \mathrm{F}$. So, there is a choice of variables for which the two sides are different, so the sides are not equivalent.

3. $p \oplus q \equiv (p \vee q) \wedge (\neg p \vee \neg q)$.

Strategy:

- To show a logical equivalence such as $p \equiv q$, we can show that $p \leftrightarrow q$ is a tautology.
- We know that $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
- Therefore, to show that $p \equiv q$, we can show that $(p \rightarrow q)$ and $(q \rightarrow p)$ are both tautologies.
- To show that $(p \rightarrow q)$ is a tautology, you **only** need to show that if $p = T$, then $q = T$
- Note that if $p = F$, the implication will always be true (so there is nothing to prove)
- To conclude, to show $p \equiv q$, you **only** need to show:
  (a) if $p = T$, then $q = T$
  (b) if $q = T$, then $p = T$

Solution:

Note that $p \oplus q = \mathrm{T}$ if and only if p and q are different. In this case, $p \vee q \equiv \neg p \vee \neg q = \mathrm{T}$ since the OR of two propositions with opposite truth values is true. Hence, if left-hand side is true, so is the right-hand side.

On the other hand, if the right-hand side is true, then p and q have opposite truth values, since otherwise either $p \vee q$ or $\neg p \vee \neg q$ is false (and hence also the entire right-hand side). This means that $p \oplus q$ is true. So if the right-hand side is true, so is the left-hand side. This means that the two sides are equivalent.

4. $\neg(p \oplus q) \equiv (\neg p) \oplus q$.

$$\begin{array}{rcll} \neg(p \oplus q) & \equiv & \neg((p \vee q) \wedge (\neg p \vee \neg q)) & \text{Exercise 3.3} \\ & \equiv & \neg(p \vee q) \vee \neg(\neg p \vee \neg q) & \text{De Morgan Laws} \\ & \equiv & (\neg p \wedge \neg q) \vee (p \wedge q) & \text{De Morgan Laws} \\ & \equiv & (\neg p \vee p) \wedge (\neg p \vee q) \wedge (\neg q \vee p) \wedge (\neg q \vee q) & \text{Distributive Laws} \\ & \equiv & (\neg p \vee q) \wedge (\neg q \vee p) & \text{Identity Laws} \\ & \equiv & (\neg p) \oplus q & \text{Exercise 3.3} \end{array}$$

5. $p \leftrightarrow q \equiv \neg(p \oplus q)$.

$$\begin{array}{rcll} p \leftrightarrow q & \equiv & (p \rightarrow q) \wedge (q \rightarrow p) & \text{Definition} \\ & \equiv & (\neg p \vee q) \wedge (p \vee \neg q) & \text{Table 7 of Rosen, Section 1.3} \\ & \equiv & (\neg p) \oplus q & \text{Exercise 3.3} \\ & \equiv & \neg(p \oplus q) & \text{Exercise 3.4} \end{array}$$

Exercise 4. *Find a compound proposition with three variables p, q, u that is*

1. *True if and only if p is true, q is false, and u is false;*
   $p \wedge \neg q \wedge \neg u$.
2. *True if and only if exactly one of the variables is true;*
   $(p \wedge \neg q \wedge \neg u) \vee (\neg p \wedge q \wedge \neg u) \vee (\neg p \wedge \neg q \wedge u)$.
3. *True if and only if at least two of the variables are true.*
   $(p \wedge q \wedge \neg u) \vee (p \wedge \neg q \wedge u) \vee (\neg p \wedge q \wedge u) \vee (p \wedge q \wedge u)$.

Exercise 5. *(Rosen, exercise 38, sec. 1.2) Five friends have access to a chat room. Is it possible to determine who is chatting if the following information is available:*

- *Either Kevin or Heather, or both, are chatting.*
- *Either Randy or Vijay, but not both, are chatting.*
- *If Abbey is chatting, so is Randy.*
- *Vijay and Kevin are either both chatting or neither is.*
- *If Heather is chatting, then so are Abbey and Kevin.*

Explain your reasoning.

Define the following propositions:

- a = "Abbey is chatting"
- h = "Heather is chatting"
- k = "Kevin is chatting"
- r = "Randy is chatting"
- v = "Vijay is chatting"

Then the description is equivalent to the following compound proposition:

$$(k \vee h) \wedge (r \oplus v) \wedge (a \to r) \wedge (v \leftrightarrow k) \wedge (h \to a \wedge k).$$

We can determine who is chatting if and only if there is a unique setting of variables for which the proposition above evaluates to true. We try to simplify the compound proposition by noting that it evaluates to true iff all the terms evaluate to true.

- The term $v \oplus r$ implies that $r \equiv \neg v$.
- The term $v \leftrightarrow k$ implies that $v \equiv k$.

So the formula simplifies to

$$(v \vee h) \wedge (a \to \neg v) \wedge (h \to a \wedge v),$$

which is much simpler since it involves only three variables.

Suppose that $h = \mathrm{T}$. Then $a \wedge v = \mathrm{T}$, since otherwise $h \to a \wedge v$ is false. Therefore $a = \mathrm{T}$ and $v = \mathrm{T}$. But then $a \to \neg v \equiv \mathrm{T} \to \mathrm{F}$ which is false, so the assumption $h = \mathrm{T}$ leads to a contradiction.

Therefore $h = \mathrm{F}$. Since $v \vee h$ is true, $v = \mathrm{T}$. Since $a \to \neg v$ is true, $a = \mathrm{F}$. The last expression is $\mathrm{F} \to \mathrm{F}$ which is true, so all the terms are true.

Therefore, there is a unique setting:

- $a = \mathrm{F}$ so Abbey is not chatting.

- $h = \text{F}$ so Heather is not chatting.
- $k = v = \text{T}$ so both Kevin and Vijay are chatting.
- $r = \neg v = \text{F}$ so Randy is not chatting.

Exercise 6. *(Rosen, exercise 36, sec. 1.2) The police have three suspects for the murder of Mr. Cooper: Mr. Smith, Mr. Jones, and Mr. Williams. Smith, Jones, and Williams each declare that they did not kill Cooper. Smith also states that Cooper was a friend of Jones and that Williams disliked him. Jones also states that he did not know Cooper and that he was out of town the day Cooper was killed. Williams also states that he saw both Smith and Jones with Cooper the day of the killing and that either Smith or Jones must have killed him. Can you determine who the murderer was if*

1. *one of the three men is guilty, the two innocent men are telling the truth, but the statements of the guilty man may or may not be true?*

   There are three possible cases:

   Case 1) When Williams is guilty, but Jones and Smith are innocent: It means Jones and Smith are telling the truth, but what William is saying may be true or may be false. Now, statements from Jones and Smith are contradicting because Smith says Cooper was a friend of Jones, but Jones says he did not know Cooper. So, It is not a possible case.

   Case 2) When Smith is guilty, but Jones and Williams are innocent: It means Jones and Williams are telling the truth, but what Smith is saying may be true or may be false. Now, statements from Jones and Williams are contradicting because Jones says he was out of town when Cooper was killed, and William says he saw Jones with Cooper. So, It is also not a possible case.

   Case 3) When Jones is guilty, but Smith and Williams are innocent: It means Smith and Williams are telling the truth, but what Jones is saying may be true or may be false. Here, nothing is contradicting from the statements of Smith and Williams. So, It is a possible case. It means Jones is guilty.

   Answer: Jones was the murderer.

2. *innocent men do not lie?*

   This is just like part (a), except that we are not told that one of the men is guilty. Can none of them be guilty? If so, then they are all telling the truth, but this is impossible because, as we just saw, some of the statements are contradictory. Can more than one of them be guilty? If, for example, they are all guilty, then their statements give us no information. So that is certainly possible.

Exercise 7. *The negation of the statement "if it rains, the ground is wet" is*

◯ *if the ground is not wet, it does not rain.*

✓ *it rains and the ground is not wet.*

◯ *if the ground is wet, it does not rain.*

◯ *if it rains, the ground is not wet.*

The proposition $A \to B$ is equivalent to $\neg A \vee B$, whose negation is $A \wedge \neg B$.

Exercise 8. *Let p and q be two propositions. Consider the two compound propositions below.*

$$((q \to p) \wedge \neg q) \to \neg p \qquad\qquad \big(((\neg q) \to (\neg p)) \wedge p\big) \to q$$

✓ *One of the compound propositions is a tautology, the other is a contingency.*

◯ *Both compound propositions are contingencies.*

◯ *One of the compound propositions is a contradiction, the other is a contingency.*

◯ *One of the compound propositions is a tautology, the other is a contradiction.*

If q is False, then $q \to p$ is True and $(q \to p) \wedge \neg q$ is True. It follows that $((q \to p) \wedge \neg q) \to \neg p$ is True if p is False, and that $((q \to p) \wedge \neg q) \to \neg p$ is False if p is True. It follows that the first statement "$((q \to p) \wedge \neg q) \to \neg p$" above is a contingency: it can be either True or False depending on the truth values of p and q. Of course, we can also use truth table 1.

p	q	$q \to p$	$\neg q$	$(q \to p) \wedge \neg q$	$\neg p$	$((q \to p) \wedge \neg q) \to \neg p$
0	0	1	1	1	1	1
0	1	0	0	0	1	1
1	0	1	1	1	0	0
1	1	1	0	0	0	1

Table 1: Truth table for the first compound proposition.

Because $(\neg q) \to (\neg p) \equiv \neg(\neg q) \leftarrow \neg(\neg p) \equiv q \leftarrow p \equiv p \to q$, the second statement "$\big(((\neg q) \to (\neg p)) \wedge p\big) \to q$" above is equivalent to "$\big((p \to q) \wedge p\big) \to q$". This latter statement is a tautology (and also known as "modus ponens", which is however not required to understand that it is a tautology): it is always True irrespective of the truth values of p and q. The same, using truth table 2.

p	q	$\neg p$	$\neg q$	$(\neg q) \to (\neg p)$	$((\neg q) \to (\neg p)) \wedge p$	$\big(((\neg q) \to (\neg p)) \wedge p\big) \to q$
0	0	1	1	1	0	1
0	1	1	0	1	0	1
1	0	0	1	0	0	1
1	1	0	0	1	1	1

Table 2: Truth table for the second compound proposition.

Exercise 9. *The compound proposition* $((\neg p \wedge q) \to (r \oplus q)) \vee (\neg s \leftrightarrow p)$ *is*

◯ *a tautology.*

✓ *a contingency.*

◯ *a contradiction.*

◯ *incorrectly formatted.*

If p and q are selected such that $\neg p \wedge q$ is true (fixing p as false and q as true), then $((\neg p \wedge q) \to (r \oplus q))$ equals $\neg r$ and $(\neg s \leftrightarrow p)$ equals s, and the expression becomes $\neg r \vee s$: this is false if r is true and s is false, and true otherwise.

Exercise 10. *For each of the following propositions, determine if it is in CNF (Conjunctive Normal Form), DNF (Disjunctive Normal Form), both, or neither.*

1. $(p \wedge q) \vee (\neg p \wedge \neg q)$

   It is a disjunction of conjunctions, so it is in DNF.

2. $(p \vee q) \wedge (\neg p \vee r)$

   It is a conjunction of disjunctions, so it is in CNF.

3. $(p \wedge q) \wedge (r \vee s)$

   It doesn't fit the structure of CNF or DNF, so it is neither.

4. $(p \vee q) \vee (r \wedge s)$

   It doesn't fit the structure of CNF or DNF, so it is neither.

5. $p \vee q$

   It can be considered as a single conjunction $(p \vee q)$, or as a disjunction of two single-term conjunctions. Thus it is both a CNF and a DNF.

6. $p \wedge q$

   It can be considered as a single disjunction $(p \wedge q)$, or as a conjunction of two single-term disjunctions. Thus it is both a CNF and a DNF.

Exercise 11. *Given the proposition: $p \to (q \wedge r)$, find:*

1. *The DNF (Disjunctive Normal Form) using a truth table.*
2. *The CNF (Conjunctive Normal Form) using a truth table.*

1. Using the truth table, we analyze the proposition when it is True.

p	q	r	$p \to (q \wedge r)$
F	F	F	T
F	F	T	T
F	T	F	T
F	T	T	T
T	F	F	F
T	F	T	F
T	T	F	F
T	T	T	T

   Upon observing the truth table, we notice that for every case where p is false (i.e., $\neg p$), the entire proposition $p \to (q \wedge r)$ is true, independently of the values of q and r. This is because a false proposition implies anything. Thus, we can directly state that whenever p is false, the proposition is true. This is why, from the truth table, we get the term $\neg p$ in our DNF. The other term (p $\wedge q \wedge r$ arises from the last row, where all propositions are true.

   'Thus,from the truth table, the DNF is:

   $$\neg p \vee (p \wedge q \wedge r)$$

2. To find the CNF, we analyze the proposition when it is False.

   From the truth table, we identify the rows where the proposition is False:

   $$p \wedge \neg q \wedge \neg r$$

   $$p \wedge \neg q \wedge r$$

   $$p \wedge q \wedge \neg r$$

   To construct our clauses, we use the negations of the variable in the rows above and connect the variables with $\vee$'s:

   $$\neg p \vee q \vee r$$

   $$\neg p \vee q \vee \neg r$$

   $$\neg p \vee \neg q \vee r$$

   Thus, we get the CNF:

   $$(\neg p \vee q \vee r) \wedge (\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee r)$$